Maya — Phase 3 Breakdown

"Feelings, an evolving relationship, and self-consistency. The moat."
6 dependency-ordered sub-phases · deterministic work front-loaded · LLM risk isolated & incremental

`3a Schema → 3b Engines → 3c Genesis → 3d Moment+Prompt → 3e Orchestrator → 3f Memory+Gate`

3a · Schema & State Models

Goal: the entire Phase-3 data layer exists and round-trips. No behavior change yet.

RISK: LOW NO LLM

Build

- **P3.1** Alembic migration `0004_phase3_state.py` — tables: `emotional_state`, `relationship_state`, `relationship_events`, `companion_commitments`; + `companions.personality` (JSONB), `companions.backstory` (TEXT)
- **models.py** — 4 new SQLAlchemy models + Companion field additions (match existing 2.0 mapped_column style)
- **P3.2** `maya/companions/templates.py` — 3 templates (flirt / devoted / best_friend)

Tests

- migration up/down clean; model insert+read round-trip; template load

Depends on: nothing. Foundation for all later sub-phases.

3b · Emotional & Relationship Engines

Goal: deterministic state logic — decay + stage transitions + commitments CRUD — fully unit-tested, visible via CLI.

RISK: LOW–MED   NO LLM

Build

- **P3.4** `maya/emotional/service.py` + `constants.py` (feeling half-lives, `decay_feeling()`). **Lazy decay on read** — computed from hours-since-`last_updated`, no scheduler.
- **P3.5** `maya/relationship/service.py` + `transitions.py` — full Appendix G table + predicates (`has_event`, `has_recent_event`, `days_since_last`…)
- **P3.6** `maya/companions/commitments.py` — CRUD only (`add`, `get_recent`); LLM extraction deferred to 3e
- **P3.11** Upgrade `maya state` CLI — show feelings, stage, intimacy/trust, events, commitments

Tests

- decay math (1.0→0.5 after one half-life); every stage transition with mock state; commitment CRUD

Depends on: 3a (tables + models).

3c · Genesis — Companion Birth

Goal: every seeded companion gets a unique backstory, initial feelings, seed commitments, and a real first message.

RISK: MEDIUM   LLM (one-shot)

Build

- **P3.3** `maya/companions/genesis.py` — Appendix E prompt → `GenesisResult`; uses `main` tier (personality matters)
- `run_genesis()` writes backstory, inits emotional + relationship state, saves seed commitments, seeds Mem0 identity, saves opening assistant message
- Wire into `maya seed` CLI

Tests

- genesis returns valid structure (mocked LLM); seed runs genesis end-to-end against test DB

Depends on: 3a (templates), 3b (emotional/relationship/commitments services), existing MemoryService.

3d · Moment Analysis & Rich Prompt

Goal: classify each moment and assemble the full Phase-3 prompt — built and tested in isolation, not yet in the live loop.

RISK: MEDIUM | LLM (fast tier)

Build

- **P3.7** `maya/conversation/moment_analyzer.py` — Appendix B, `fast` tier, <500ms; invalid JSON/timeout → default `chitchat` (never blocks)
- **P3.9** Rich `prompt_builder.py` — Appendix C template + Appendix H moment guidance; identity / feelings / relationship / moment / memories / commitments / recent / message blocks
- `count_tokens()` via `tiktoken` — fail loud if > 8000 tokens (new dep)

Tests

- moment analyzer parse + fallback path; prompt assembly snapshot; token-budget guard trips

Depends on: 3b (emotional/relationship DTOs), 3a (commitments model).

3e · Orchestrator Integration

Goal: wire everything into the live chat loop — the behavioral leap. Highest-risk sub-phase.

RISK: HIGH | LLM (multi-call)

Build

- **P3.10** Upgrade `Orchestrator.handle_message` — parallel context gather (memory, emotional, relationship, commitments, recent), moment analysis, rich prompt, main LLM call
- **Post-processing** (awaited *after* reply sent): Mem0 extract, emotional update, relationship increment, **P3.6** LLM commitment extraction, event logging, stage-transition eval
- **DebugOrchestrator refactor** — orchestrator emits step events via optional callback; web debug panel subscribes. *One* code path — kills the 3/10 duplication bug class.

Tests

- full `handle_message` with mocked LLM; post-processing applies all state updates; debug callback fires expected events

Depends on: 3b, 3c, 3d (everything converges here).

3f · Memory Quality & Gate

Goal: richer memories + full Phase-3 gate verification. Ship-readiness.

RISK: LOW–MED   LLM (config)

Build

- **P3.8** `maya/memory/prompts.py` — Appendix A companion-aware extraction prompt; inject into `build_mem0_config` (`custom_fact_extraction_prompt` , `custom_update_memory_prompt`)
- Run full P3 gate (below) + cost check

Tests / Gate

- qualitative memory richness check; all gate items below pass

Depends on: 3e (live loop) for end-to-end validation.

Phase 3 Exit Gate

Gate item	Verified in
`maya seed` produces unique backstory + first message	3c
Emotional state changes between calm vs high-intensity exchanges	3e
Relationship stage transitions correctly (tested)	3b (logic) · 3e (live)
After 24h test-clock, feelings have decayed	3b
No self-contradiction across 20-message identity conversation	3e + 3f
`maya state` shows rich, accurate snapshot	3b (built) · 3e (populated)
Custom Mem0 prompts richer than defaults (qualitative)	3f
All Phase 1 + Phase 2 tests still pass	every sub-phase (regression)
LLM cost per turn average < $0.10	3f

Why This Slicing

- **Risk front-loading** — 3a/3b are pure deterministic, fully unit-testable, zero LLM cost. Confidence built before any model is called.
- **Incremental LLM exposure** — one-shot genesis (3c) → isolated prompt/analysis (3d) → full loop (3e). Each LLM surface validated before the next stacks on it.
- **Duplication killed once** — the DebugOrchestrator refactor lands exactly where the orchestration grows complex (3e), not before.

- **Every P3.x sub-part mapped exactly once** — P3.1→3a, P3.2→3a, P3.3→3c, P3.4→3b, P3.5→3b, P3.6→3b+3e, P3.7→3d, P3.8→3f, P3.9→3d, P3.10→3e, P3.11→3b.